

# Random Variables as Functions, Expected Value and Variance

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).

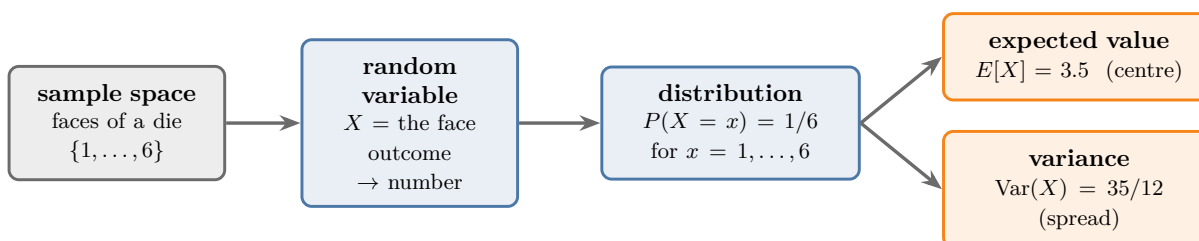


- **Builds on:** Random variables ([Note 240](#)).
- **Leads to:** Poisson distribution ([Note 560](#)).
- **Compare with:** Measures of central tendency ([Note 221](#)).

## 1. Overview

**Key point:** A random variable turns each outcome into a number; its distribution then gives two summary numbers: the expected value (where the values centre) and the variance (how far they spread).

### From outcomes to a centre and a spread



**Figure 1:** From the sample space of a die to its expected value and variance

Figure 1 shows the path this Note follows for one die. The faces form the sample space; a random variable  $X$  writes each face as a number; the distribution gives each number's probability. From the distribution we compute the **expected value**  $E[X] = 3.5$ , the centre, and the **variance**  $\text{Var}(X) = 35/12$ , the spread.

Random variables, discrete and continuous ones, and probability distributions are introduced in the [random variables and distributions Note](#). Here we look closer at what a random variable is, then compute its mean and variance.

## 2. A random variable is a function

**Key point:** Despite its name, a random variable is not a variable but a function: its input is an outcome from the sample space, its output is a real number, and the rule in between is set by the event we want to study.

The name “random variable” is misleading. A random variable is a **function**: a rule that takes an input and turns it into an output. In the context of probability:

Stated as a definition: a random variable is a function that maps each outcome of a random experiment (each member of the sample space) to a real number.

To understand any function we need three things: its input, its output and its rule.

- **Input:** one outcome from the sample space. Every outcome is fed in, one at a time.
- **Output:** a real number, one for each outcome.
- **Rule:** decided by the event we want to study (section 2.2).

Random variables get capital letters such as  $X$  or  $Y$ ; the [random variables and distributions Note](#) gives this convention, and “the set of possible values” view used there is the list of this function’s outputs.

## 2.1 Simple rules: coin and die

**Key point:** For a coin we can map head to 1 and tail to 0; for a die we can map each face to its own number.

**Tossing a coin.** The sample space is  $\{H, T\}$ . A random variable  $X$  can map head to 1 and tail to 0:

$$X(H) = 1, \quad X(T) = 0$$

**Rolling a die.** The sample space is  $\{1, 2, 3, 4, 5, 6\}$ . A random variable  $Y$  can map each face to itself:  $Y(1) = 1$ ,  $Y(2) = 2$ , and so on up to  $Y(6) = 6$ .

These rules look like a change of names and nothing more. Their use shows once the experiment has more complex outcomes, such as pairs.

## 2.2 The rule comes from the event

**Key point:** Rolling two dice gives 36 pairs; the event “sum is 7” calls for the rule “add the dice”, and the event “die 1 beats die 2” calls for “subtract die 2 from die 1”.

Roll two dice together. The sample space has 36 outcomes, the pairs  $(1, 1), (1, 2), \dots, (6, 6)$  (laid out as a grid in the [conditional probability Note](#)). What number we attach to each pair depends on the question.

**Event: “the sum is 7”.** The natural rule is to add the two dice:

$$X(d_1, d_2) = d_1 + d_2$$

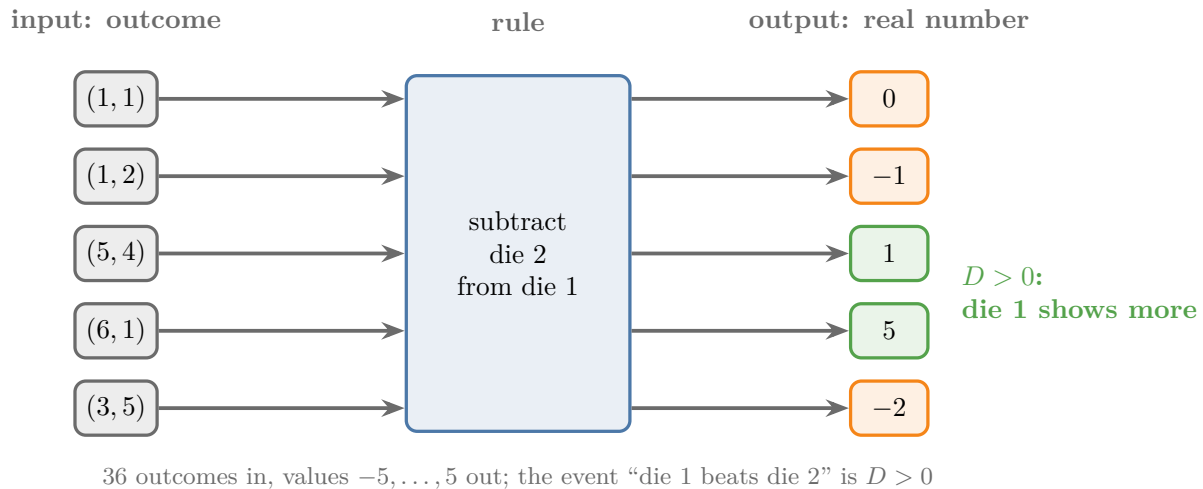
So  $(3, 4) \rightarrow 7$ ,  $(1, 3) \rightarrow 4$ ,  $(6, 6) \rightarrow 12$ . The event is now simply  $X = 7$ , which happens for 6 of the 36 pairs. The full distribution of  $X$ , from 2 to 12, is the table in the [random variables and distributions Note](#) (section 3.1).

**Event: “die 1 shows more than die 2”.** Now the useful rule is the difference:

$$D(d_1, d_2) = d_1 - d_2$$

Figure 2 feeds five pairs through this rule. The output is positive when die 1 is higher, 0 when they are equal and negative when die 2 is higher. So the event “die 1 beats die 2” is  $D > 0$ .

**$D = \text{die 1} - \text{die 2}$ : one outcome in, one number out**



**Figure 2:** The random variable  $D$  maps each pair of dice to the difference die 1 minus die 2

Counting the 36 pairs for each value of  $D$  gives its distribution ( $P$  is  $P(D = d)$ ):

| $d$   | -5             | -4             | -3             | -2             | -1             | 0              | 1              | 2              | 3              | 4              | 5              |
|-------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| pairs | 1              | 2              | 3              | 4              | 5              | 6              | 5              | 4              | 3              | 2              | 1              |
| $P$   | $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{3}{36}$ | $\frac{4}{36}$ | $\frac{5}{36}$ | $\frac{6}{36}$ | $\frac{5}{36}$ | $\frac{4}{36}$ | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ |

$$P(D > 0) = \frac{5 + 4 + 3 + 2 + 1}{36} = \frac{15}{36} = \frac{5}{12} \approx 0.417$$

The same sample space gave two different random variables, because we asked two different questions. Turning outcomes into numbers is what lets us tabulate, plot and calculate with them: the distribution tables, PMFs and PDFs of the [random variables and distributions](#) [Note](#) all start here.

**Python:** A random variable as a Python function.

```
import itertools
from collections import Counter

# the 36 pairs of two dice
S = list(itertools.product(range(1, 7), repeat=2))
```

```
def difference(outcome):          # the random variable D
    return outcome[0] - outcome[1]

# count the pairs for each value of D
counts = Counter(difference(o) for o in S)
itertools.product(range(1, 7), repeat=2) lists every ordered pair of faces.
Counter counts how often each output appears; dividing by 36 gives the table above.
The Notebook (notebook.ipynb) runs every calculation in this Note.
```

A random variable whose outputs are separate values, as here, is discrete; one whose outputs fill a range, such as a student's CGPA, is continuous. Both kinds are covered in the [random variables and distributions Note](#). This Note works with discrete random variables.

### 3. Expected value

**Key point:** The expected value  $E[X]$  is the long-run average of a random variable: each possible value times its probability, summed. For a die it is 3.5.

The **mean of a random variable**, usually called its **expected value**, is the average outcome of a random process repeated many times. More technically, it is a weighted average of the possible values, each weighted by its probability.

#### 3.1 The mean as a weighted average

**Key point:** The ordinary mean can be rewritten as each distinct value times its share of the data; the expected value uses probabilities in place of shares.

Take the six numbers 5, 3, 4, 5, 3, 3. Their mean (see the [measures of central tendency Note](#)) is

$$\bar{x} = \frac{5 + 3 + 4 + 5 + 3 + 3}{6} = \frac{23}{6} \approx 3.833$$

The same sum can be grouped by value: 5 appears twice, 4 once and 3 three times.

$$\bar{x} = \frac{2 \times 5 + 1 \times 4 + 3 \times 3}{6} = \frac{2}{6} \times 5 + \frac{1}{6} \times 4 + \frac{3}{6} \times 3 = \frac{23}{6}$$

The second form says: take each **distinct** value and multiply it by the share of the data it makes up. A share of the data is an empirical probability (see the [empirical and theoretical probability Note](#)). Replacing the shares by true probabilities gives the expected value.

#### 3.2 The formula

**Key point:**  $E[X] = \sum x_i P(X = x_i)$ , summed over the possible values.

1. **In words:** multiply every possible value of the random variable by its probability, and add the products.

2. **Formula:** for a random variable with  $n$  possible values  $x_1, \dots, x_n$ ,

$$E[X] = \sum_{i=1}^n x_i P(X = x_i)$$

The expected value is also written  $\mu$  (mu).

3. **Example:** one die, each face with probability  $1/6$ :

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{21}{6} = 3.5$$

So if we roll a die many times and average the faces, the average will be close to 3.5. No single roll can show 3.5: the expected value is a long-run average, not a value we expect to see on any one trial.

More examples with the same formula:

| Random variable | Values and probabilities           | $E[X]$                                            |
|-----------------|------------------------------------|---------------------------------------------------|
| Coin, head = 1  | 1 and 0, each $1/2$                | $1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = 0.5$ |
| Sum of two dice | 2 to 12 (table in Note 240)        | $252/36 = 7$                                      |
| Difference $D$  | $-5$ to $5$ (table in section 2.2) | 0                                                 |

The difference has expected value 0 because the table is symmetric: each positive value is balanced by a negative one with the same probability. Neither die is favoured.

**Extra:** For a coin with head = 1,  $E[X] = P(\text{head})$ . This holds for any Bernoulli variable (see the [PMF Note](#)):  $E[X] = 1 \cdot p + 0 \cdot (1 - p) = p$ . The average of 0/1 values is the share of 1s, which is why `tosses.mean()` gave the share of heads in the [empirical and theoretical probability Note](#).

### 3.3 Checking by simulation

**Key point:** The average of 100,000 simulated die rolls is 3.4998, within 0.0002 of the expected value 3.5.

**Python:** The long-run average of a die.

```
import numpy as np
```

```
rng = np.random.default_rng(42)          # fixed seed
rolls = rng.integers(1, 7, 100_000)      # 100,000 rolls, 1 to 6
rolls.mean()                             # 3.49982
```

The more rolls, the closer the average gets to 3.5, for the same reason empirical probabilities settle on theoretical ones. The Notebook also plots the running average after each roll.

The result is not a coincidence: the expected value is exactly the number such averages settle on.

**Extra:** For a continuous random variable, the sum becomes an integral over the PDF  $f(x)$  (see the [PDF Note](#)):  $E[X] = \int x f(x) dx$ . The idea is the same: each value weighted by how likely it is.

## 4. Variance of a random variable

**Key point:** The variance  $\text{Var}(X)$  is the expected squared distance of the random variable from its expected value; for a die it is  $35/12 \approx 2.92$ .

The variance of a column of data is the average squared distance of the values from their mean (see the [measures of dispersion Note](#)). The variance of a random variable is built the same way, with the expected value taking the place of every average.

### 4.1 From the data formula to the random-variable formula

**Key point:** In the data formula, replace the mean by  $E[X]$  and the averaging by another expected value:  $\text{Var}(X) = E[(X - E[X])^2]$ .

For data, the (population) variance is

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

It does three things: subtract the mean from each value, square each distance, and average the squares. For a random variable:

- the mean  $\bar{x}$  becomes the expected value  $E[X]$ ;
- each value  $x_i$  becomes the random variable  $X$  itself, so the squared distance is the random quantity  $(X - E[X])^2$ ;
- “average the squares” becomes “take the expected value of the squares”.

1. **In words:** the variance is the expected value of the squared distance between  $X$  and its expected value.

2. **Formula:**

$$\text{Var}(X) = E[(X - E[X])^2] = \sum_{i=1}^n (x_i - \mu)^2 P(X = x_i)$$

3. **Example:** one die,  $\mu = 3.5$ . The squared distances of the six faces are

$$(1 - 3.5)^2 = 6.25, (2 - 3.5)^2 = 2.25, (3 - 3.5)^2 = 0.25, 0.25, 2.25, 6.25$$

Each has probability  $1/6$ :

$$\text{Var}(X) = \frac{6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25}{6} = \frac{17.5}{6} = \frac{35}{12} \approx 2.917$$

The **standard deviation** of  $X$  is the square root,  $\sqrt{35/12} \approx 1.708$ , back in the units of  $X$ .

Weighting by probabilities matches the data formula that divides by  $n$ , not by  $n - 1$ . The  $n - 1$  version estimates a population’s variance from a sample; a random variable’s distribution already describes the whole population.

## 4.2 The shortcut formula

**Key point:**  $\text{Var}(X) = E[X^2] - (E[X])^2$ : the expected value of the square minus the square of the expected value.

Expanding the square gives a second formula that is often quicker.

1. **In words:** take the expected value of  $X^2$ , then subtract the square of the expected value of  $X$ .
2. **Formula:**

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

3. **Example:** one die.

$$E[X^2] = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6} \approx 15.167$$

$$\text{Var}(X) = \frac{91}{6} - 3.5^2 = \frac{91}{6} - \frac{49}{4} = \frac{182 - 147}{12} = \frac{35}{12}$$

The same value as the definition.

The derivation uses three rules for expected values. They hold for any random variables, and need no independence (Grinstead and Snell, *Introduction to Probability*, 2nd ed., 1997, section 6.1, Theorem 6.2, which notes that expectations add whether or not the summands are independent):

- **Constants:** a constant  $c$  has  $E[c] = c$ . It takes one value with probability 1, so its average is itself.  $E[X]$  is a constant too, a single number such as 3.5.
- **Scaling:**  $E[cX] = c E[X]$ . Multiplying every value by  $c$  multiplies the weighted average by  $c$ .
- **Sums:**  $E[X + Y] = E[X] + E[Y]$ , the **linearity of expectation**.

Write  $\mu = E[X]$  and expand the square  $(X - \mu)^2 = X^2 - 2\mu X + \mu^2$ :

$$\text{Var}(X) = E[X^2 - 2\mu X + \mu^2]$$

Split the sum (sums rule) and take the constants  $2\mu$  and  $\mu^2$  out (scaling and constants rules):

$$\text{Var}(X) = E[X^2] - 2\mu E[X] + \mu^2$$

Since  $E[X] = \mu$ , the middle term is  $2\mu^2$ :

$$\text{Var}(X) = E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2$$

Both formulas are used constantly, for discrete and continuous random variables alike.

**Extra:** The middle step is sometimes justified by calling  $2X$  and  $E[X]$  “independent”. Independence is not what is used:  $E[X]$  is a constant, and constants come out of an expected value by the scaling rule. Independence matters for  $E[XY] = E[X]E[Y]$

(Grinstead and Snell, Theorem 6.4), which is not needed here. Grinstead and Snell derive the shortcut exactly this way, as Theorem 6.6 of section 6.2.

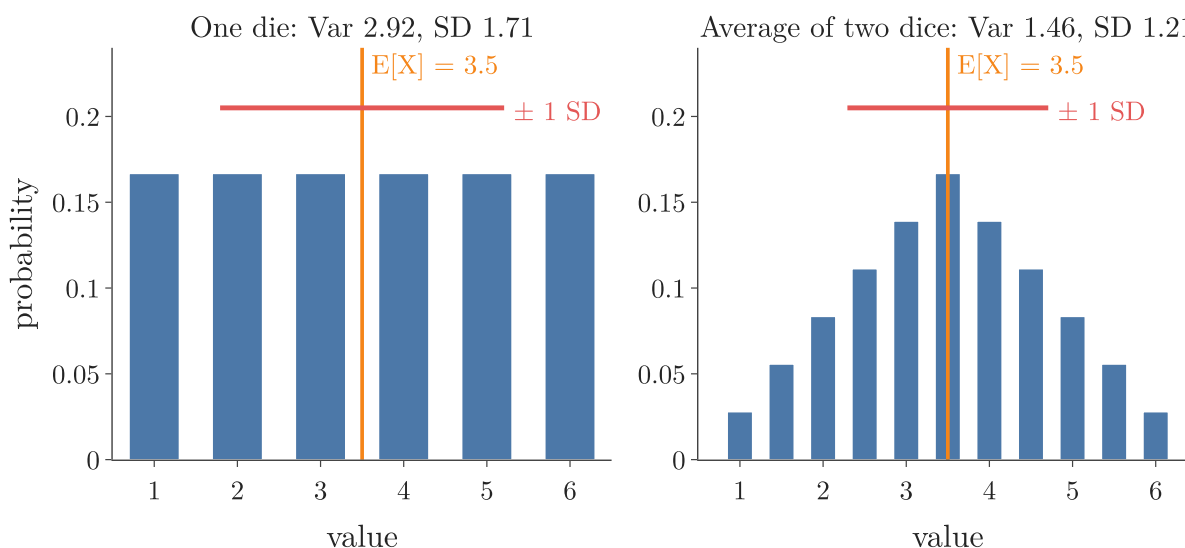
**Python:** Both formulas, and the simulated rolls.

```
faces = np.arange(1, 7)
p = np.full(6, 1 / 6)
mu = (faces * p).sum() # 3.5
var_def = ((faces - mu) ** 2 * p).sum() # 2.9167
var_short = (faces ** 2 * p).sum() - mu ** 2 # 2.9167
rolls.var() # 2.9173
rolls.var() divides by n (NumPy's default); on 100,000 rolls it lands within 0.001 of 35/12.
```

### 4.3 Same mean, different spread

**Key point:** Two random variables can share an expected value and differ in variance; the variance tells how far single outcomes stray from the centre.

Figure 3 compares one die with the **average** of two dice,  $(d_1 + d_2)/2$ . Both have expected value 3.5. The single die is equally likely to land anywhere from 1 to 6, while the average piles up near 3.5: extreme values need both dice to be extreme.



**Figure 3:** One die and the average of two dice: same expected value, different variance

| Random variable     | $E[X]$ | $\text{Var}(X)$       | Standard deviation |
|---------------------|--------|-----------------------|--------------------|
| One die             | 3.5    | $35/12 \approx 2.917$ | 1.708              |
| Average of two dice | 3.5    | $35/24 \approx 1.458$ | 1.208              |

The expected value says where the outcomes centre; the variance says how much a single outcome can be trusted to land near it.

**Extra:** Averaging two independent dice halved the variance, from  $35/12$  to  $35/24$ . In general, the average of  $n$  independent copies has variance  $\sigma^2/n$  (Grinstead and Snell, Theorem 6.9). With  $n = 2$  and  $\sigma^2 = 35/12$  this gives  $35/24$ , as in the table. The models averaged in [bagging](#) are not independent, since they are trained on overlapping data. If each pair has correlation  $\rho$ , the average of  $B$  of them has variance  $\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$  (Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, 2nd ed., 2009, section 15.2, equation 15.1). Averaging still lowers the variance, but only the second term shrinks as  $B$  grows.

## 5. Summary

| Quantity              | Formula                           | One die                 | Coin (head = 1)                    |
|-----------------------|-----------------------------------|-------------------------|------------------------------------|
| Random variable       | outcome $\rightarrow$ real number | face $\rightarrow$ face | $H \rightarrow 1, T \rightarrow 0$ |
| Expected value        | $E[X] = \sum x_i P(X = x_i)$      | 3.5                     | 0.5                                |
| Variance (definition) | $E[(X - E[X])^2]$                 | $35/12 \approx 2.917$   | 0.25                               |
| Variance (shortcut)   | $E[X^2] - (E[X])^2$               | $91/6 - 49/4 = 35/12$   | $0.5 - 0.25 = 0.25$                |
| Standard deviation    | $\sqrt{\text{Var}(X)}$            | 1.708                   | 0.5                                |

- A random variable is a function from outcomes to real numbers; the event we study decides its rule.
- One sample space can carry many random variables: sum, difference, larger face.
- The expected value is a probability-weighted average, the long-run mean of many trials; it need not be a possible value.
- The variance is the expected squared distance from the expected value; the shortcut  $E[X^2] - (E[X])^2$  gives the same number.
- Expected values are linear: constants come out, sums split. No independence is needed for this.

## 6. Key terms

| Term                                    | Meaning                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| Expected value $E[X]$                   | The probability-weighted average of a random variable's values; its long-run mean; also written $\mu$ |
| Mean of a random variable               | Another name for its expected value                                                                   |
| Variance of a random variable           | $\text{Var}(X) = E[(X - E[X])^2]$ : the expected squared distance from the expected value             |
| Shortcut variance formula               | $\text{Var}(X) = E[X^2] - (E[X])^2$                                                                   |
| Standard deviation of a random variable | The square root of its variance, in the units of $X$                                                  |
| Linearity of expectation                | $E[aX + bY + c] = a E[X] + b E[Y] + c$ , for any random variables                                     |